Assignment: SQL Notebook for Peer Assignment

Estimated time needed: **60** minutes.

Introduction

Using this Python notebook you will:

1. Understand the Spacex DataSet
2. Load the dataset into the corresponding table in a Db2 database
3. Execute SQL queries to answer assignment questions

Overview of the DataSet

SpaceX has gained worldwide attention for a series of historic milestones.

It is the only private company ever to return a spacecraft from low-earth orbit, which it first accomplished in December 2010. SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars wheras other providers cost upward of 165 million dollars each, much of the savings is because Space X can reuse the first stage.

Therefore if we can determine if the first stage will land, we can determine the cost of a launch.

This information can be used if an alternate company wants to bid against SpaceX for a rocket launch.

This dataset includes a record for each payload carried during a SpaceX mission into outer space.

Download the datasets

This assignment requires you to load the spacex dataset.

In many cases the dataset to be analyzed is available as a .CSV (comma separated values) file, perhaps on the internet. Click on the link below to download and save the dataset (.CSV file):

Spacex DataSet

In [1]:
```
!pip install sqlalchemy==1.3.9
```

```
Collecting sqlalchemy==1.3.9
  Downloading SQLAlchemy-1.3.9.tar.gz (6.0 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 6.0/6.0 MB 65.3 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... one
Building wheels for collected packages: sqlalchemy
  Building wheel for sqlalchemy (setup.py) ...done
  Created wheel for sqlalchemy: filename=SQLAlchemy-1.3.9-cp312-cp312-linux_x86_6
4.whl size=1160111 sha256=d3993f5359cd7a8885ecf034e2e9ffb7a438d5afc24c4937684be9a
5e98df674
  Stored in directory: /home/jupyterlab/.cache/pip/wheels/b3/1c/42/0e26b8d512adc6
bce10ff71a05229366b4ccec641cd3b42111
Successfully built sqlalchemy
Installing collected packages: sqlalchemy
  Attempting uninstall: sqlalchemy
    Found existing installation: SQLAlchemy 2.0.37
    Uninstalling SQLAlchemy-2.0.37:
      Successfully uninstalled SQLAlchemy-2.0.37
ERROR: pip's dependency resolver does not currently take into account all the pac
kages that are installed. This behaviour is the source of the following dependenc
y conflicts.
jupyterhub 5.2.1 requires SQLAlchemy>=1.4.1, but you have sqlalchemy 1.3.9 which
is incompatible.
Successfully installed sqlalchemy-1.3.9
```

Connect to the database

Let us first load the SQL extension and establish a connection with the database

In [1]:
```
!pip install ipython-sql
!pip install ipython-sql prettytable
```

```
Collecting ipython-sql
  Downloading ipython_sql-0.5.0-py3-none-any.whl.metadata (17 kB)
Collecting prettytable (from ipython-sql)
  Downloading prettytable-3.17.0-py3-none-any.whl.metadata (34 kB)
Requirement already satisfied: ipython in /opt/conda/lib/python3.12/site-packages
(from ipython-sql) (8.31.0)
Requirement already satisfied: sqlalchemy>=2.0 in /opt/conda/lib/python3.12/site-
packages (from ipython-sql) (2.0.37)
Collecting sqlparse (from ipython-sql)
  Downloading sqlparse-0.5.5-py3-none-any.whl.metadata (4.7 kB)
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om ipython-sql) (1.17.0)
Requirement already satisfied: ipython-genutils in /opt/conda/lib/python3.12/site
-packages (from ipython-sql) (0.2.0)
Requirement already satisfied: greenlet!=0.4.17 in /opt/conda/lib/python3.12/site
-packages (from sqlalchemy>=2.0->ipython-sql) (3.1.1)
Requirement already satisfied: typing-extensions>=4.6.0 in /opt/conda/lib/python
3.12/site-packages (from sqlalchemy>=2.0->ipython-sql) (4.12.2)
Requirement already satisfied: decorator in /opt/conda/lib/python3.12/site-packag
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Requirement already satisfied: matplotlib-inline in /opt/conda/lib/python3.12/sit
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Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in /opt/conda/lib/py
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Requirement already satisfied: pygments>=2.4.0 in /opt/conda/lib/python3.12/site-
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Downloading prettytable-3.17.0-py3-none-any.whl (34 kB)
Downloading sqlparse-0.5.5-py3-none-any.whl (46 kB)
Installing collected packages: sqlparse, prettytable, ipython-sql
Successfully installed ipython-sql-0.5.0 prettytable-3.17.0 sqlparse-0.5.5
Requirement already satisfied: ipython-sql in /opt/conda/lib/python3.12/site-pack
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Requirement already satisfied: pexpect>4.3 in /opt/conda/lib/python3.12/site-pack
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Requirement already satisfied: pygments>=2.4.0 in /opt/conda/lib/python3.12/site-
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-packages (from stack_data->ipython->ipython-sql) (3.0.0)
Requirement already satisfied: pure_eval in /opt/conda/lib/python3.12/site-packag
es (from stack_data->ipython->ipython-sql) (0.2.3)
```

In [5]:
```
!pip install ipython-sql
```

```
Collecting ipython-sql
  Downloading ipython_sql-0.5.0-py3-none-any.whl.metadata (17 kB)
Collecting prettytable (from ipython-sql)
  Downloading prettytable-3.17.0-py3-none-any.whl.metadata (34 kB)
Requirement already satisfied: ipython in /opt/conda/lib/python3.12/site-packages
(from ipython-sql) (8.31.0)
Collecting sqlalchemy>=2.0 (from ipython-sql)
  Downloading sqlalchemy-2.0.48-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x
86_64.manylinux_2_28_x86_64.whl.metadata (9.5 kB)
Collecting sqlparse (from ipython-sql)
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-packages (from ipython-sql) (0.2.0)
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Requirement already satisfied: pygments>=2.4.0 in /opt/conda/lib/python3.12/site-
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Requirement already satisfied: traitlets>=5.13.0 in /opt/conda/lib/python3.12/sit
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Requirement already satisfied: wcwidth in /opt/conda/lib/python3.12/site-packages
(from prettytable->ipython-sql) (0.2.13)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in /opt/conda/lib/python3.12/s
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Requirement already satisfied: ptyprocess>=0.5 in /opt/conda/lib/python3.12/site-
packages (from pexpect>4.3->ipython->ipython-sql) (0.7.0)
Requirement already satisfied: executing>=1.2.0 in /opt/conda/lib/python3.12/site
-packages (from stack_data->ipython->ipython-sql) (2.1.0)
Requirement already satisfied: asttokens>=2.1.0 in /opt/conda/lib/python3.12/site
-packages (from stack_data->ipython->ipython-sql) (3.0.0)
Requirement already satisfied: pure_eval in /opt/conda/lib/python3.12/site-packag
es (from stack_data->ipython->ipython-sql) (0.2.3)
Downloading ipython_sql-0.5.0-py3-none-any.whl (20 kB)
Downloading sqlalchemy-2.0.48-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86
_64.manylinux_2_28_x86_64.whl (3.3 MB)
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Downloading prettytable-3.17.0-py3-none-any.whl (34 kB)
Downloading sqlparse-0.5.5-py3-none-any.whl (46 kB)
Installing collected packages: sqlparse, sqlalchemy, prettytable, ipython-sql
  Attempting uninstall: sqlalchemy
    Found existing installation: SQLAlchemy 1.3.9
    Uninstalling SQLAlchemy-1.3.9:
      Successfully uninstalled SQLAlchemy-1.3.9
Successfully installed ipython-sql-0.5.0 prettytable-3.17.0 sqlalchemy-2.0.48 sql
parse-0.5.5
```

In [6]:
```
%load_ext sql
```

In [7]:
```
import csv, sqlite3
import prettytable
prettytable.DEFAULT = 'DEFAULT'

con = sqlite3.connect("my_data1.db")
cur = con.cursor()
```

In [8]:
```
!pip install -q pandas
```

In [9]:
```
%sql sqlite:///my_data1.db
```

In [10]:
```
import pandas as pd
df = pd.read_csv("https://cf-courses-data.s3.us.cloud-object-storage.appdomain.c
df.to_sql("SPACEXTBL", con, if_exists='replace', index=False,method="multi")
```

Out[10]: 101

Note:This below code is added to remove blank rows from table

In [11]:
```
df.head()
```

Out[11]:

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Or
0	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	L
1	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of...	0	L (I
2	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	L (I
3	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	L (I
4	2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	L (I

In [12]:
```
%sql create table SPACEXTABLE as select * from SPACEXTBL where Date is not null
```

```
 * sqlite:///my_data1.db
(sqlite3.OperationalError) table SPACEXTABLE already exists
[SQL: create table SPACEXTABLE as select * from SPACEXTBL where Date is not null]
(Background on this error at: https://sqlalche.me/e/20/e3q8)
```

Tasks

Now write and execute SQL queries to solve the assignment tasks.

Note: If the column names are in mixed case enclose it in double quotes For Example "Landing_Outcome"

Task 1

Display the names of the unique launch sites in the space mission

In [13]:
```
%%sql
SELECT DISTINCT Launch_Site
FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[13]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Task 2

Display 5 records where launch sites begin with the string 'CCA'

In [14]:
```
%%sql
SELECT *
FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[14]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

In [15]:

```
%%sql
SELECT SUM(Payload_Mass__kg_) AS Total_Payload_Mass
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[15]:

Total_Payload_Mass
45596

Task 4

Display average payload mass carried by booster version F9 v1.1

In [16]:

```
%%sql
SELECT AVG(Payload_Mass__kg_) AS Avg_Payload
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[16]:

Avg_Payload
2928.4

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

In [17]:
```
%%sql
SELECT MIN(Date) AS First_Success_Landing
FROM SPACEXTABLE
WHERE Landing_Outcome LIKE 'Success%';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[17]:

First_Success_Landing
2015-12-22

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

In [18]:
```
%%sql
SELECT Booster_Version, Payload_Mass__kg_, Landing_Outcome
FROM SPACEXTABLE
WHERE Landing_Outcome LIKE 'Success%'
AND Payload_Mass__kg_ BETWEEN 4000 AND 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[18]:

Booster_Version	PAYLOAD_MASS__KG_	Landing_Outcome
F9 FT B1022	4696	Success (drone ship)
F9 FT B1026	4600	Success (drone ship)
F9 FT B1021.2	5300	Success (drone ship)
F9 FT B1032.1	5300	Success (ground pad)
F9 B4 B1040.1	4990	Success (ground pad)
F9 FT B1031.2	5200	Success (drone ship)
F9 B4 B1043.1	5000	Success (ground pad)
F9 B5 B1046.2	5800	Success
F9 B5 B1047.2	5300	Success
F9 B5 B1046.3	4000	Success
F9 B5 B1048.3	4850	Success
F9 B5 B1051.2	4200	Success
F9 B5B1060.1	4311	Success
F9 B5 B1058.2	5500	Success
F9 B5B1062.1	4311	Success

Task 7

List the total number of successful and failure mission outcomes

In [19]:

```
%%sql
SELECT Mission_Outcome, COUNT(*) AS Total_Count
FROM SPACEXTABLE
GROUP BY Mission_Outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[19]:

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

In [20]:

```
%%sql
SELECT Booster_Version
FROM SPACEXTABLE
```

```
WHERE Payload_Mass__kg_ = (
    SELECT MAX(Payload_Mass__kg_)
    FROM SPACEXTABLE
);
```

```
 * sqlite:///my_data1.db
Done.
```

Out[20]:

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

In [22]:

```
%%sql
SELECT
    substr(Date,6,2) AS Month,
    Landing_Outcome,
    Booster_Version,
    Launch_Site
FROM SPACEXTABLE
WHERE substr(Date,1,4) = '2015'
AND Landing_Outcome LIKE '%Failure%'
AND Landing_Outcome LIKE '%drone ship%';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[22]:

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

In [21]:
```
%%sql
SELECT Landing_Outcome,
COUNT(*) AS Outcome_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Outcome_Count DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[21]:

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Reference Links

- Hands-on Lab : String Patterns, Sorting and Grouping
- Hands-on Lab: Built-in functions
- Hands-on Lab : Sub-queries and Nested SELECT Statements
- Hands-on Tutorial: Accessing Databases with SQL magic
- Hands-on Lab: Analyzing a real World Data Set

Author(s)

Lakshmi Holla

Other Contributors

Rav Ahuja

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